

1 Abstract

2 Urban green spaces (UGS) are critical for maintaining biodiversity and supporting wildlife,
3 especially in heavily urbanized environments. In temperate regions, winter brings significant
4 challenges for resident bird species, which can be either compounded or alleviated in urban
5 regions. ~~We~~ To better understand how birds use urban habitats during winter, we experimentally
6 translocated Northern Cardinals (*Cardinalis cardinalis*) among sites spanning an urbanization
7 gradient in Montreal, Canada, and used VHF telemetry to quantify their home range sizes, scale
8 of size, habitat selection, across spatial scales and habitat associations in *Cardinalis cardinalis*
9 (*Northern Cardinal*) translocated across an urban gradient in Montreal, Canada, over three
10 winters. *C. cardinalis* populations increased across the northeastern United States and
11 southeastern Canada in the past 100 years with most *C. cardinalis* restricted to cities,
12 likely due to the presence of bird feeders. *C. cardinalis* Cardinals in the most urbanized study site
13 had larger home ranges compared to birds wintering in the suburban and peri-urban sites. Habitat
14 selection occurred at the smallest scales measurable (25-50 m) across all sites
15 (urban/suburban/peri-urban) implying very fine-scale selection. This emphasises the importance
16 of microhabitat features such as vegetation structure and bird feeder presence. Habitat
17 associations varied across the urban gradient, except for proximity to bird feeders which was
18 consistently related to *C. cardinalis* habitat use by cardinals. In less urbanized areas, birds
19 selected for vertically complex vegetation, likely because *C. cardinalis* select shrubby,
20 edge habitat. In more urbanized areas, birds were more likely to use areas with a greater density
21 of roads, implying *C. cardinalis* select the most residential areas. *C. cardinalis* Cardinals
22 exhibited behavioral plasticity in response to urbanization, underscoring the need for locally
23 tailored conservation strategies and suggesting that fine-scale ecological understanding to design

24 biodiversity-friendly UGS that support wildlife and human well-being can help guide locally
25 tailored green space management aimed at supporting urban-adapted species year-round.

26 *Keywords:* behavioral plasticity, fine-scale movement, habitat association, home range size,
27 manual VHF telemetry, *Cardinalis cardinalis* non-breeding period, scale of habitat selection,
28 urbanization gradient, winter

29 Lay summary

- 30 • Urban green spaces (UGS) can support help wildlife survive in urbanized areas, yet cities,
31 but little is known about birds' how birds use of UGS these spaces during the winter
32 months in temperate regions northern climates.
- 33 • We used tracked Northern Cardinals in Montreal, Canada, using small radio
34 tag transmitters to track follow where they move during winter movements of *Cardinalis*
35 *cardinalis* (Northern Cardinal) across an urban gradient.
- 36 • We By tracking these birds, we quantified the area how much space they used through and
37 what kinds of places they favoured in areas ranging from greener neighbourhoods to
38 denser parts of the city.
- 39 • Cardinals spending the winter, the scale of habitat selection, and the association to
40 variables related to nature and human presence in the most built-up areas used larger
41 territories and relied on resources found close by, such as vegetation features and human-
42 provided food sources like bird feeders.
- 43 • *C. cardinalis* had larger home ranges in the most urban site, habitat selection occurred at
44 a small scale across all sites and association to human or natural variables varied
45 depending on the urbanization level.

- *C. cardinalis* displayed behavioral plasticity, highlighting the importance of adapting conservation strategies to local urban contexts.
- Overall, cardinals showed a strong ability to adjust to city life, suggesting that well-designed urban green spaces can support urban-adapted birds year-round.

Introduction

Urbanization is accelerating worldwide (Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022)(Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022), altering landscapes and bringing novel challenges to wildlife including habitat fragmentation, habitat loss, and chronic pollution (Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023)(Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023). Future global urban expansion could lead to an additional 11-33 million hectares of habitat loss by 2100, with a disproportionately large increase in habitat fragmentation (Li et al. 2022)(Li et al. 2022). This human-altered landscape habitat loss and fragmentation can affect in turn reshape wildlife spatial ecology, including patterns of home range size and movement patterns. A meta-analysis showed both birds and mammals had smaller home ranges in urban areas and that urbanization restricts spatial movements for most animals (O'Donnell and delBarco-Trillo 2020). During the breeding period, the communication distance of *Zonotrichia leucophrys* (White-crowned Sparrows) in the San-Francisco Bay was highly predictive of its territory size, highlighting that in the urban landscape, noise pollution may drive songbirds into smaller home ranges in the breeding season (Simberloff et al. 2023). Birds in urban environments also tend to be central foragers, associated with a particular location, doing some back-and-forth between this central area and other foraging locations (Hahs et al. 2023). We highlight that this applies for the breeding season, while research in avian urban ecology remains heavily biased towards studies

69 conducted during this period. However, there are exceptions. *Passer domesticus* (House
70 Sparrows) in urban locations in South Bohemia, Czech Republic, had larger foraging areas and
71 flew greater distances to get food (Havlíček et al. 2022). *P. domesticus* selected sites such as
72 storage areas or poultry holding, which are both associated with easier food, of poor quality
73 (Havlíček et al. 2022). Thus, natural food source for sparrows are scarce and scattered in the
74 urban landscape, driving the sparrows to forage in larger territories (Havlíček et al.
75 2022). Nonetheless, most research on urban birds occurs in the breeding season, and there is far
76 less in winter, and those studies that do exist contrast with breeding season studies. This
77 knowledge gap is important because birds often exhibit different behaviours outside of the
78 breeding season. For example, *Cardinalis cardinalis* (Northern Cardinals) form winter flocks
79 (while vigorously defending their territories against conspecifics during the breeding season) and
80 alter their patterns of space use during the non-breeding season in Kentucky (Ritchison and
81 Omer 1990). Consequently, factors influencing habitat use during winter may differ substantially
82 from those operating during the breeding season.

83 Birds respond to these urban pressures through behavioural adjustments, reflecting both
84 challenges and opportunities, including increased predation pressure, heightened disturbances,
85 novel anthropogenic resources and human-made structures. Habitat use in the city can differ
86 significantly from that in rural environments, even within a single species. These differences can
87 reflect hierarchical habitat selection (Johnson 1980), as animals select habitat at multiple spatial
88 scales—from a geographic range (first order), to the home range of an individual (second order),
89 to the selection of specific patches for daily activities (third and fourth orders). In the urban
90 context these decisions are influenced by novel pressures such as human disturbances, artificial
91 structures, and altered resource availability. Resource Selection Functions (RSFs) offer a

92 quantitative method of assessing shifting habitat preferences along an urban gradient by
 93 modelling the probability of use of a specific habitat compared to habitat availability (Manly et
 94 al. 2002; Poirier et al. 2024). Birds adapt to urban habitats in response to challenges and
 95 opportunities, including increased predation pressures, heightened disturbances, novel
 96 anthropogenic resources, and human-made structures. To compensate for disturbances from
 97 noise, light and human activity, some birds may change their nesting behaviour—for example by
 98 nesting higher in the canopy (e.g., *Pica pica* (Eurasian Magpie); Šálek et al. 2020) or using off-
 99 ground artificial surfaces increasing fitness (e.g., *Junco hyemalis* (Dark-eyed ~~Junco~~
 100 Bressler et al., 2020). Urban habitats can also provide foraging opportunities which can be
 101 exploited by some species like *Falco peregrinus* (Peregrine ~~Falco~~
 102 shown to use buildings as perches and nesting sites, capitalizing on disoriented migratory birds
 103 wandering near lit glass structures (Kauffman et al. 2003; DeCandido and Allen 2006; Altwegg
 104 et al. 2014)(Kauffman et al. 2003; DeCandido and Allen 2006; Altwegg et al. 2014). Although
 105 some species can adapt and thrive in urban areas, urban tolerance varies across species and
 106 across annual cycles, with urban tolerance peaking during the winter months (potentially due to
 107 the urban heat island effect) (Callaghan et al. 2021).
 108 Habitat use in the city can differ significantly from that in rural environments, even within a
 109 single species. These differences can reflect hierarchical habitat selection (Johnson 1980), as
 110 animals select habitat at multiple spatial scales—from geographic range (first order), to the home
 111 range of an individual (second order), to the selection of specific patches for daily activities
 112 (third and fourth orders). In the urban context those decisions are influenced by novel pressures
 113 such as human disturbances, artificial structures, and altered resource availability. Quantifying
 114 habitat use by comparing used locations with what is available in the habitat can provide

115 [valuable insights into how wildlife responds to urbanization and identify habitat features that are](#)
116 [more strongly associated with both occurrence and space use \(Manly et al. 2002; Poirier et al.](#)
117 [2024\).](#)

118 Urban birds may outperform rural birds during ~~the~~ winter ~~season~~, when natural resources are less
119 available and weather is harsher. Cities can provide a more stable wintering
120 ~~environments~~[environment](#) for bird species that use ~~bird~~ feeders or that are generalists or
121 omnivores (i.e., corvids), which exploit anthropogenic resources ~~through the~~ winter (~~Jokimäki~~
122 ~~et al. 2022~~)([Jokimäki et al. 2022](#)). In other cases, cities can serve as refuge during periods of
123 extreme weather, as local colonization of cities by birds increased during the colder conditions
124 due to cities acting as heat islands (Latimer and Zuckerberg 2021). Nonetheless, occupancy
125 dynamics in temperate regions during winter are highly species-specific, with thermal tolerance
126 playing an important role in determining which species would be able to persist in urban
127 environments during this season. Moreover, energy demands for small resident passerines
128 increase with latitude in winter, meaning that ~~the~~ birds spending their winter in regions such as
129 southern Quebec, must consume significantly more energy to maintain their body temperatures
130 than those wintering farther south—which is especially challenging given the scarcity of
131 resources during the cold months (Fristoe 2015). Despite these physiological and environmental
132 challenges, along with the potential added effects of urbanization, winter remains an
133 underrepresented season in experimental, ecological research (Dinh et al. 2023). This gap in
134 knowledge highlights the importance of investigating how birds are occupying urban habitats in
135 winter, particularly in temperate regions where winter survival is a major limiting factor for
136 resident bird species.

137 To address this gap, we studied *Cardinalis cardinalis* (Northern Cardinal) across an urban
138 gradient in one of Canada's largest cities, Montreal, Quebec. ~~C. cardinalis~~Cardinals are large
139 songbirds (weighing ~~between~~ 40 to 45 grams), which allows for ~~the~~ deployment of radio tags
140 capable of lasting almost the entirety of the non-breeding season. ~~These birds are also very well~~
141 ~~adapted to urban life, able to thrive and persist across a wide range of urbanization gradients,~~
142 ~~from suburban neighbourhoods to densely populated areas of the city. They are resident in~~
143 ~~southern Quebec, a region of extreme cold and loss of natural seeds during the cold winter~~
144 ~~months.~~ In the last 100 years, ~~C. cardinalis~~cardinals expanded their distribution range
145 northwards, and have been anecdotally closely associated to urban and suburban landscapes in
146 those areas (Beddall 1963; Smith et al. 2011; Robert et al. 2019). ~~We tracked weekly movements~~
147 ~~of C. cardinalis across an urbanization gradient in the city of Montreal, Quebec, Canada. This~~
148 ~~study has four objectives: (i) investigate how C. cardinalis'~~They are able to thrive and persist
149 ~~across a wide range of urbanization gradients, from suburban neighbourhoods to densely~~
150 ~~populated areas of the city. Cardinals are now resident in southern Quebec, a region of extreme~~
151 ~~cold (sometimes reaching -27 °C with wind chill; Environment and Climate Change Canada) and~~
152 ~~limited natural food sources during the winter months (Government of Canada 2004; Turcotte~~
153 ~~and Desrochers 2005). As part of a translocation experiment, we captured and released~~
154 ~~individuals at sites distributed along this urbanization gradient in Montreal to evaluated how they~~
155 ~~are using such habitat. We tracked weekly movements to assess how the birds adjust their~~
156 ~~movements and habitat use after being displaced into environments with varying levels of~~
157 ~~urbanization.~~
158 This study has four objectives: (i) investigate how cardinals' home range size varies across an
159 urbanization gradient through the winter season, (ii) quantify the scale at which C.

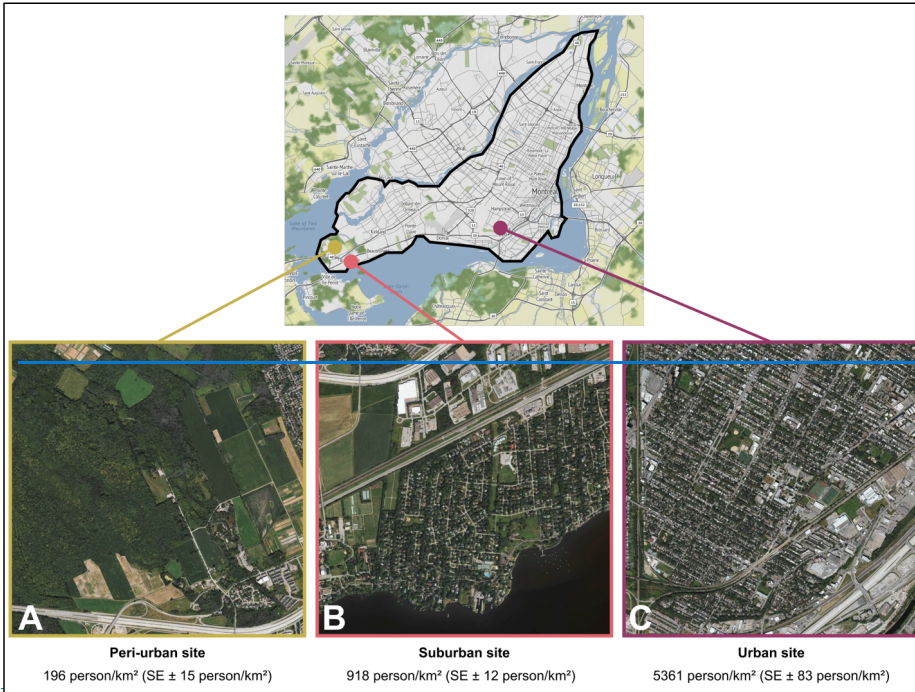
160 *cardinaliscardinals* are associated with certain factors of their habitat, (iii) ~~analyse *C. cardinalis*'~~
161 ~~association~~determine the relative strength of anthropogenic and natural vegetation to
162 ~~different~~predict habitat ~~factors~~suse by ~~comparing used and available habitats~~cardinals during the
163 non-breeding season, and (iv) document the spring movements from non-breeding to breeding
164 grounds. We predicted that wintering *C. cardinaliscardinals* require larger home ranges (both
165 core and total) and select habitats at larger scales when in dense urban locations due to food
166 sources being more scattered (Havlíček et al., 2022). ~~In addition, we predicted that *C. cardinalis*~~
167 ~~would have a higher probability of presenece in locations with bird feeders, lower population~~
168 ~~density, and including a complex vertical vegetation structure (Rousseau et al. 2015; Carraseo et~~
169 ~~al. 2019). Finally, we predicted that *C. cardinalis*(Havlíček et al. 2022). In addition, we predicted~~
170 that locations with bird feeders, lower population density and a complex vertical vegetation
171 structure would be associated with a higher probability of cardinals. Finally, we predicted that
172 cardinals would transition from their breeding sites in forested/canopy covered areas towards
173 residential areas in the fall to take advantage of bird feeders during winter, reversing the
174 movement in spring.

175 **Materials and methods**

176 Study locations

177 The island of Montreal is located in southern Quebec, Canada, with a population of ~2 million
178 people coupled with high population density. The city's climate is comprised of four distinct
179 seasons with cold winters (temperatures ~~often reaching -20 °C~~) and ~~warm summers (with~~
180 ~~temperatures sometimes exceeding 30 °C)~~. ~~This research was conducted in can range between -23~~
181 ~~°C and 4 °C during the month of January)~~ and warm summers (~~with temperatures sometimes~~
182 ~~exceeding 30 °C; WeatherSpark)~~. This research was conducted at three study sites (Figure 1)

183 with varying levels of urbanization. Each site includes two release locations randomly assigned
184 to the individuals. The peri-urban site is located on the western end of the island of Montreal at
185 the McGill Bird Observatory (MBO; 45°25'51.4"N 73°56'18.5"W). The MBO is part of the 30
186 km² ~~of the~~ *Grand parc de l'Ouest*. The suburban site is located in the Town of Baie d'Urfé
187 (45°24'39.2"N 73°55'50.0"W). The urban study site is located in the Notre-Dame-de-Grâce
188 neighbourhood of the city (45°27'35.7"N 73°38'30.2"W). Because [*C. cardinalis*cardinals](#) can be
189 found in all three sites, we assumed that the translocated birds would be able to survive and
190 thrive at each location.



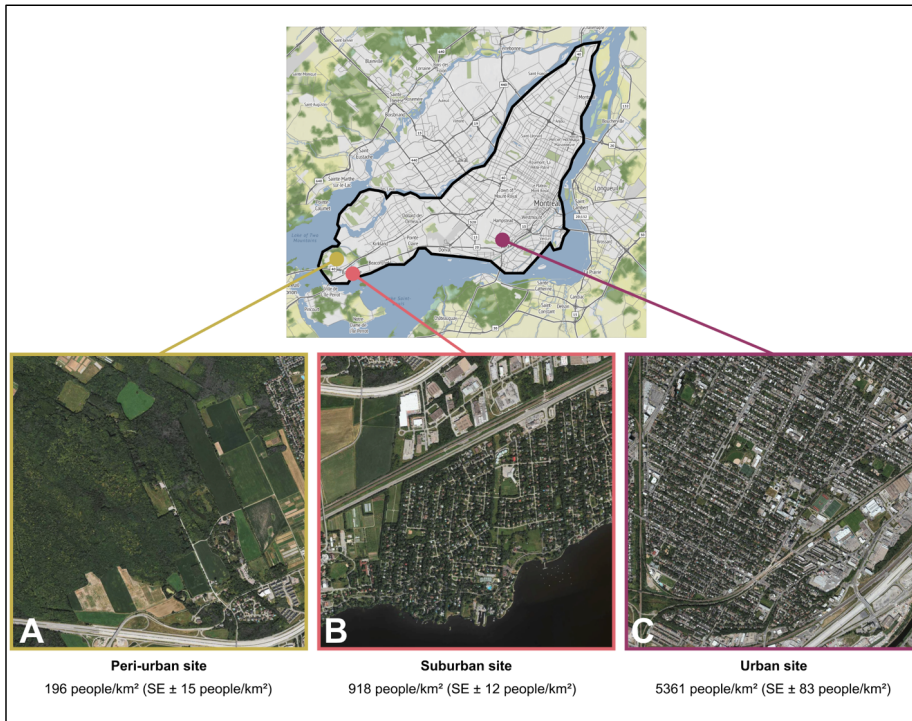


Figure 1. Study locations on the Island of Montreal A) The peri-urban site is at the McGill Bird Observatory (2% of the area covered by streets and/or roads). B) The suburban site (5% of the area covered by streets and/or roads) is located in the residential Town of Baie d’Urfé. C) The urban site (13% of the area covered by streets and/or roads) is located in Notre-Dame-de-Grâce neighbourhood.

The three translocation sites had differed significantly in their level of urbanization from one another, following a Kruskal-Wallis test and a Bonferroni-corrected pairwise Wilcoxon test ($p < 0.001$) based on population density data from the Cencensus Program R package (von Bergmann et al. 2021), using census data from the Government of Canada for the year 2021. All

three sites are characterized by relatively high levels of green space compared to the broader urban context of Montreal. The gradient used here ranges from the higher percentages of canopy covers in the peri-urban site to the median canopy cover for the Montreal area (~ 26%) (HealthyDesign.City 2021). As such, this gradient does not fully capture the extent of urbanization present across the entire city and should be interpreted as a localized urbanization gradient. We did try translocating *C. cardinalis* to [areasan area](#) with lower canopy cover early on, but due to two immediate mortalities we decided to omit this site.

Radio tagging, banding & translocation

All birds were caught during routine fall migration monitoring at MBO and were fitted with colour bands and Motus NanoTags following standard banding and morphological measurements (see Supplemental Information). *C. cardinalis* translocated to sites away from the peri-urban location (suburban and urban), as well as control birds held at the peri-urban site, underwent a holding time of 45 minutes before release to standardize treatment amongst individuals. For the winter of the year 2022-2023 (winter 1), we were able to consistently track 20 birds released as control at the peri-urban site, which served as a pilot winter to assess the feasibility of the project. During the winter of 2023-2024 (winter 2), ~~a total of 45 birds were~~ tagged and banded [45 birds](#), with 15 individuals translocated to the urban site, 15 translocated to the suburban site and 15 released as controls at the peri-urban site. In 2024-2025 (winter 3), ~~a total of 42 birds were~~ tagged and banded [42 birds](#), with 5 individuals released as controls at the peri-urban site, 16 individuals translocated ~~atto~~ the suburban site and 21 individuals translocated ~~atto~~ the urban site. During the reproductive season, *C. cardinalis* with tags (weighing 1.6 g [double the 0.8 g of ~~our~~ the tags [used in the present study](#)]) attached defended their nests less vigorously, but did not change their provisioning effects (Barron et al. 2013). *C.*

226 *cardinalis* Cardinal nests with tagged parents also had lower daily survival rates and were less
227 likely to fledge offspring (Barron et al. 2013). However, these tag effects were relatively modest
228 and observed across diverse habitats, making it improbable that they would only manifest in our
229 dense urban site while remaining undetected in peri-urban and suburban locations. Birds were
230 banded under Animal Use Protocol 2007-5447 from McGill University and federal banding
231 permits 10973, 10973B, 10973D issued by the Canadian Wildlife Service, and radio-tagged
232 under Animal Use Protocol 22BF01 and 23BF01 from Environment and Climate Change
233 Canada.

234 *C. cardinalis* Cardinals' abundance and seasonal distribution

235 ~~To investigate *C. cardinalis*' historical abundance in Quebec, Canada, we used data from~~
236 ~~the Audubon Christmas Bird Count, more specifically we extracted the number of *C. cardinalis*~~
237 ~~recorded per party hours in the states of Maine, New York, New Hampshire and Vermont (United~~
238 ~~States), along with the data from Quebec (Canada) (National Audubon Society 2015).~~
239 We used the number of cardinals recorded per party hour in Audubon Christmas Bird Count data
240 to compare the historical abundance of cardinals in Quebec, Canada, against the states of Maine,
241 New York, New Hampshire and Vermont (United States) (National Audubon Society 2020).

242 Non-breeding season tracking and home range estimations

243 The birds were monitored using an array of automated Motus towers and handheld telemetry
244 receivers. We conducted manual radio tracking of each individual two to three times per week
245 between the months of October and May each year. The areas monitored were approximately ~5
246 km² (500 ha) at the peri-urban site, ~6 km² (600 ha) for the suburban site and ~18 km² (1800 ha)
247 for the urban site. Tracking effort and spatial coverage differed among sites due to logistical
248 constraints associated with weather conditions and the structure of the landscape. At the peri-

249 [urban site, the absence of road networks and lower levels of signal interference allowed for the](#)
250 [tracking team to move more freely and locate the individuals more rapidly, resulting in a smaller](#)
251 [monitored area. In contrast, tracking the birds in suburban and urban areas meant that the](#)
252 [tracking team's movements were constrained by road access, built infrastructure, private land](#)
253 [and higher levels of radio interference. In addition, individuals released at the suburban and](#)
254 [urban sites often moved farther away from their release location, requiring the tracking team to](#)
255 [search across larger areas to maintain contact with tagged birds. These differences in spatial](#)
256 [coverage reflect practical limitations of tracking birds in the heterogeneous landscape of urban](#)
257 [environments and should be considered when comparing movement patterns across sites.](#) The
258 people tracking the birds by hand (trackers) ~~would cover as much ground as possible at each~~
259 ~~study location with the goal of re-locating~~ [aimed to record](#) at least one GPS location for each
260 individual per week using a Garmin ETREX 22x GPS. [Trackers would typically spend 5 hours](#)
261 [tracking per session, twice per week, and started by first tracking down newly relocated birds](#)
262 [and/or birds with known recent locations. Once these individuals are found, trackers would move](#)
263 [on to expand their search areas to try and find missing individuals.](#) Details on the tracking
264 method used and estimations of precision can be found in the Supplemental Information section.
265 For all three study sites, we calculated the average home range size using the Kernel Density
266 Estimation (KDE), ~~a non-parametric way to estimate~~. [The home range derived from the](#)
267 ~~probability density function utilization~~ [distribution \(UD\) represents the minimum area containing](#)
268 [a specified proportion of a random variable based on a finite data sample](#) ~~an individual's space~~
269 [use](#) (in the present case, the ~~*C. cardinalis*~~ [cardinals'](#) GPS location recorded in the field) with [the](#)
270 [kernelUD function from](#) the adehabitatHR: Home Range Estimation R package version 0.4.22
271 ~~(Calenge 2024);~~ [\(Worton 1989; Calenge 2023, 2024\).](#) For example, the 95% home range

represents the smallest area in which there is a 95% probability of finding the individual. We selected the 50% and 95% contours to investigate respectively the core area used within the home range of the birds and the broader home range excluding the most extreme locations (outliers). A minimum of 5 GPS locations were required to adequately run the function to calculate the KDEs meaning that individuals with fewer points than this threshold were excluded. Between 5 and 48 points per individual were used to estimate the KDEs, with the average number of points per bird being 17-18.

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}$$

Where:

$K(u)$ is the kernel function evaluated at u , providing the weight assigned to a data point based on its distance to the target location.

$u = \frac{x - x_i}{h}$ is the standardized distance between the location x (where the density is being estimated) and the observed data point x_i , scaled by the bandwidth h .

h is the bandwidth parameter, which determines the smoothness of the resulting density estimate; smaller values of h lead to a rougher estimate, while larger values produce a smoother result.

$\frac{1}{\sqrt{2\pi}}$ is the normalization constant ensuring that the kernel integrates to one.

e is Euler's number (approximately 2.718), the base of the natural logarithm.

Scale-of-effect

We first identified the scale-of-effect (i.e., the spatial extent at which landscape variables have the strongest relationship with a species' response') by extracting variables at four spatial scales (25, 50, 100, 200 meters; Martin & Fahrig, 2012). These spatial scales were chosen because they cover a wide range of possible scale-of-

294 effect that are [highly](#)-relevant in urban ecology, considering that habitat heterogeneity is scale-
 295 dependent. [However, because scale-of-effect analyses identify the spatial extent that best predicts](#)
 296 [responses to the tested range, these values should be interpreted as indicating generally fine or](#)
 297 [coarse spatial responses rather than precise ecological thresholds.](#) For each scale we ran
 298 generalized linear mixed models (GLMMs) at each study location using all five predictor
 299 variables as well as year, site, and individual ID as random effects. We ranked models using
 300 Akaike Information Criterion (AICc) values, which take into account small sample sizes
 301 (Burnham and Anderson 2002). The best model was identified as having the lowest AICc value.
 302 We also calculated the ΔAICc to evaluate competing models following Burnham and Anderson
 303 (2002) (models with a $\Delta\text{AICc} < 2$ were considered to have substantial support, competing with
 304 the best model and models with a $\Delta\text{AICc} > 10$ were deemed to have no support, not competing
 305 with the best model). [Once the scale-of-We acknowledge that low per-individual sample sizes](#)
 306 [might reduce the precision of the scale-of-effect estimates, as models fitted on few observations](#)
 307 [per individual could be susceptible individual-level variation driving apparent scale preferences.](#)
 308 [To try mitigating this, we included the individuals' IDs as random effect, and we interpret scale-](#)
 309 [of-effect results at the population level rather than an individual level habitat selection.](#)
 310 [Because the four buffer sizes represent nested spatial extents, we expect the variables extracted](#)
 311 [from these buffers to be positively correlated. We assessed cross-scale correlations for each](#)
 312 [predictor type using Pearson's "r". Correlations were highest for population density \(\$r = 0.96-\$](#)
 313 [0.99 across all sites\), moderate for mean vegetation heights \(\$r = 0.37-0.94\$ \) and road length \(\$r =\$](#)
 314 [0.32-0.95\), and lowest for SD vegetation height \(\$r = 0.33-0.84\$ \), decreasing with increasing scale](#)
 315 [separation in all cases. Variables from different scales were never included within the same](#)
 316 [model—each candidate model uses a single buffer size—so cross-scale correlations do not](#)

317 [inflate within-model multicollinearity \(all VIF < 10\). The buffer comparison therefore constitutes](#)
318 [a scale-selection exercise rather than a multi-scale additive approach. Once the scale-of-effect](#)
319 was determined for each location, we conducted a resource selection function.

320 Resource Selection Function

321 We compared 'used' vs. 'available' habitat using a resource selection function for [C.](#)
322 ~~[cardinalis](#)~~[cardinals](#) during the non-breeding season and across an urban gradient. We used
323 environmental and anthropogenic variables extracted from available spatial data as well that
324 collected in the field. This included mean and standard deviation (SD) of vegetation height (1m
325 resolution) from OpenForest (Government and Municipalities of Quebec 2016), distance to
326 nearest bird feeder (collected through systematic field observations), population density data
327 (census unit) using the Cancensus R package (von Bergmann et al. 2021) and spatial data on the
328 streets and road from the island of Montreal (Ville de Montréal 2018). Unused locations were
329 derived at a 1:1 ratio by creating random locations within 300 meters of the used location, at a
330 random angle. ~~The average number of true observations for each individual ranges from 5 to 48~~
331 ~~with an average number of points of 18. Having a ratio of 1:1, the same numbers apply for the~~
332 ~~random observations. For each study site, we ran a~~[We used the matched case-control design](#)
333 [elaborated by Fortin and have applied a 300 meters buffer to reflect the median daily bird](#)
334 [movement \(Fortin et al. 2005, 2009\). Because this is a population-level resource selection](#)
335 [function, in which individual IDs are included as random effects and used-available pairs from](#)
336 [all individuals are evaluated together, there is no required minimum number of locations per](#)
337 [individual, as it would be the case for an individual-level RSF. The average number of true](#)
338 [observations for each individual ranges from 5 to 48 with an average number of points of 18.](#)
339 [Having a ratio of 1:1, the same numbers apply for the random observations. The data extracted](#)

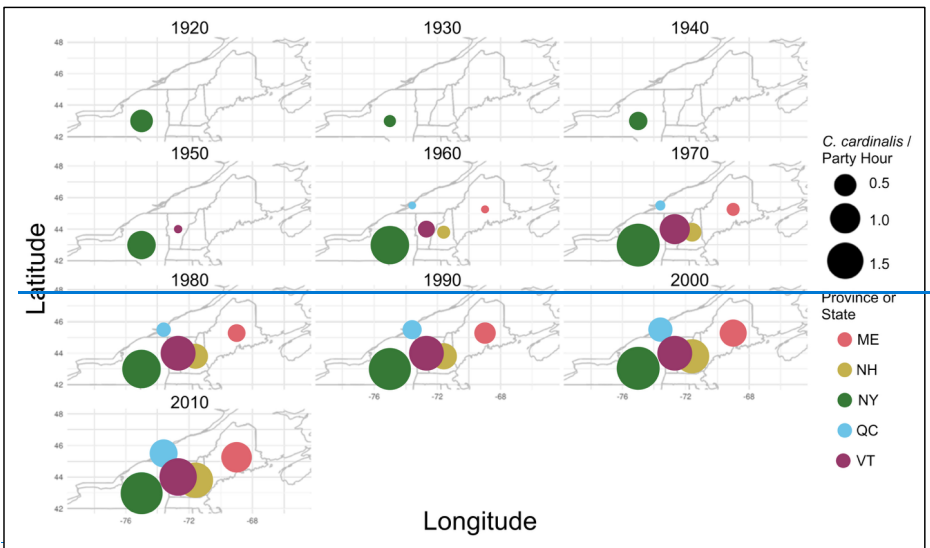
340 was scaled and centered before running the models. We built our candidate model set to compare
 341 the relative strengths of anthropogenic and natural vegetation variables to predict cardinal habitat
 342 use. This included four *a priori* hypothesis-based models including, (1) the “Full model” (Used ~
 343 Mean vegetation height + Sd vegetation height + Population density + Road length + Distance to
 344 nearest feeder + (1|Winter) + (1|Site) + (1|ID)), including all of the variables, plus the winter
 345 (years of sampling), relocation site and bird ID as random effects. ~~The second model we ran was~~
 346 ~~to account for the additive importance of anthropogenic and natural vegetation variables.~~ (2) the
 347 “Vegetation height” model (Used ~ Mean vegetation height + (1|Winter) + (1|Site) + (1|ID)),
 348 assuming the vertical height of vegetation was ~~a primary factor for decision making.~~ ~~The third~~
 349 ~~model we ran was the dominant variable related to habitat use by cardinals.~~ (3) the “Sd
 350 vegetation height” model (Used ~ Sd vegetation height + (1|Winter) + (1|Site) + (1|ID)), ~~in this~~
 351 ~~case, we assume assuming that the cardinal habitat use is related to~~ vertical vegetation
 352 composition ~~was a more predominant decision-making factor compared to height on its own.~~ ~~The~~
 353 ~~fourth, and (4) the “Anthropogenic” model we ran was the “Human model”~~ (Used ~ Population
 354 density + Road length + Distance to nearest feeders + (1|Winter) + (1|Site) + (1|ID)), which
 355 assumed that ~~the C. cardinal cardinal habitat use~~ would ~~rely mostly on human-be~~ related cues to
 356 ~~select anthropogenic variable~~. We also included a ~~wintering habitat.~~ ~~The final model we ran was~~
 357 ~~the “Null” model, including only the with~~ random effects ~~variables only~~ (Used ~ + (1|Winter) +
 358 (1|Site) + (1|ID)). ~~We then once again selected the best~~ ~~We ranked all five~~ models according to
 359 the corrected Akaike Information Criterion (AICc) following the method previously described by
 360 Burnham and Anderson (2002). We used Variance Inflation Factor (VIF) to test for
 361 multicollinearity (variables with VIF values > 10 were considered to have problematic
 362 collinearity and were removed) among predictor variables and Hosmer-Lemeshow as a

goodness-of-fit test (Hosmer and Lemeshow 1980). VIFs ranged between 1.05 and 3.31 across all predictor variables. All statistical analysis were performed using Program R version 4.4.2 (R Core Team 2024).

Results

Description of *C. cardinalis* trends across years and seasons

In southern Quebec, the *C. cardinalis* population has seen a steep increase in abundance since the 1970s (Fig. 2). As the years progress, the number of *C. cardinalis* reported in the northeastern United States increased as the species propagated northward, reaching southern Quebec in the 1960s (Fig. 2).



Historical and seasonal context of Cardinals in southern Quebec

To place our winter telemetry results in the context of the species' recent expansion into southern Quebec and its seasonal dynamics within the study region, we summarized long-term regional

abundance and local banding records. In southern Quebec, cardinals have increased in abundance since the 1970s, reflecting the species' well-documented northward range expansion from the northeastern United States into southern Quebec beginning in the 1960s (Figure 2).

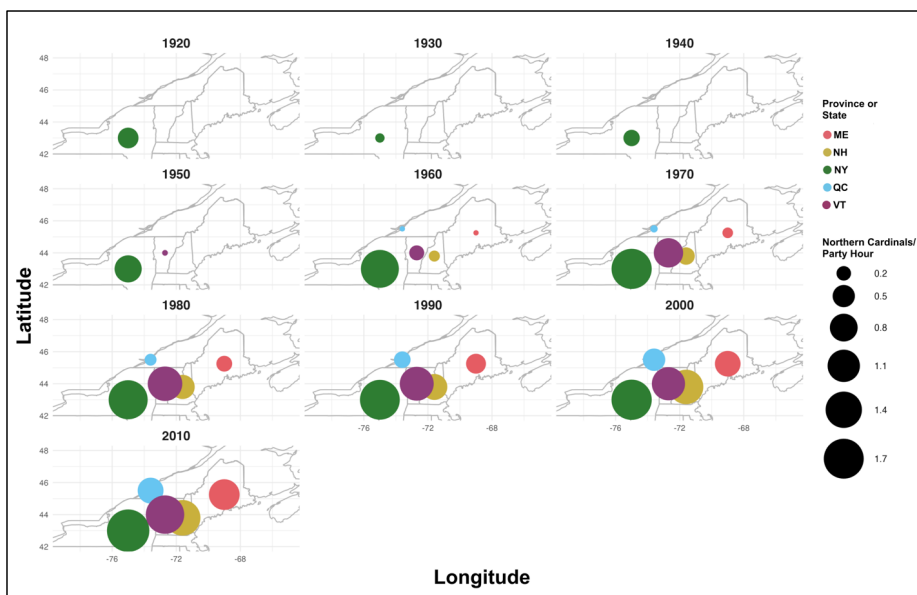
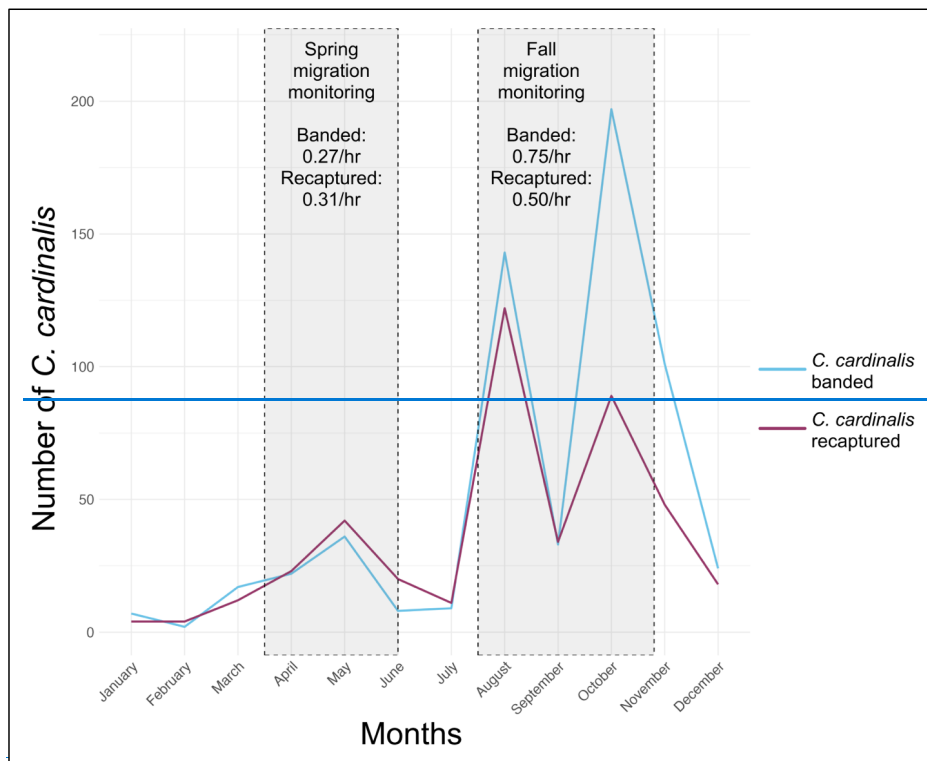


Figure 2. Number of *G. cardinalis* observations per party hour between the 1920s and the 2010s according to state (United States of America) or province (Canada) as part of the Audubon Christmas Bird Count. *G. cardinalis* captures at Nineteen years of banding data from the McGill Bird Observatory over a 19-year period peak first in May (both banded and recaptured birds; also revealed distinct seasonal patterns in captures (Figure 3), approximately at the middle point of the spring migration). Capture rates were higher during the fall monitoring program. During the

entirety of (~0.75 birds/h) than during the spring program, *C. cardinalis* are banded at a rate of
~0.27 individual per hour of mist-net opening and *C. cardinalis* are recaptured at a
slightly higher rate of ~0.31 individual per hour of mist-net opening. When the fall migration
monitoring program occurs, there is a steep increase in the number of *C. cardinalis* banded
and recaptured. Banding rate goes up to ~0.75 individual per hour of mist-net opening
(almost three times the banding rate during the spring), while recapture rate increases to
0.50 individual per hour of mist-net opening. Thus, there is a greater proportion of unbanded
individuals, indicating an influx of new birds in late fall. The individuals captured in the
fall migration monitoring program are mostly aged as juveniles or unknowns (Supplementary
Material: Figure S1B, C). Despite this, we can still identify two distinct peaks of adult
C. cardinalis on site (specifically in mid-May and in late July; Supplementary
Material: Figure S1A) implying that this movement involved both post-fledging
dispersal and adult movement from breeding to non-breeding sites. Local movements between
breeding and wintering areas contribute to seasonal variation in captures.



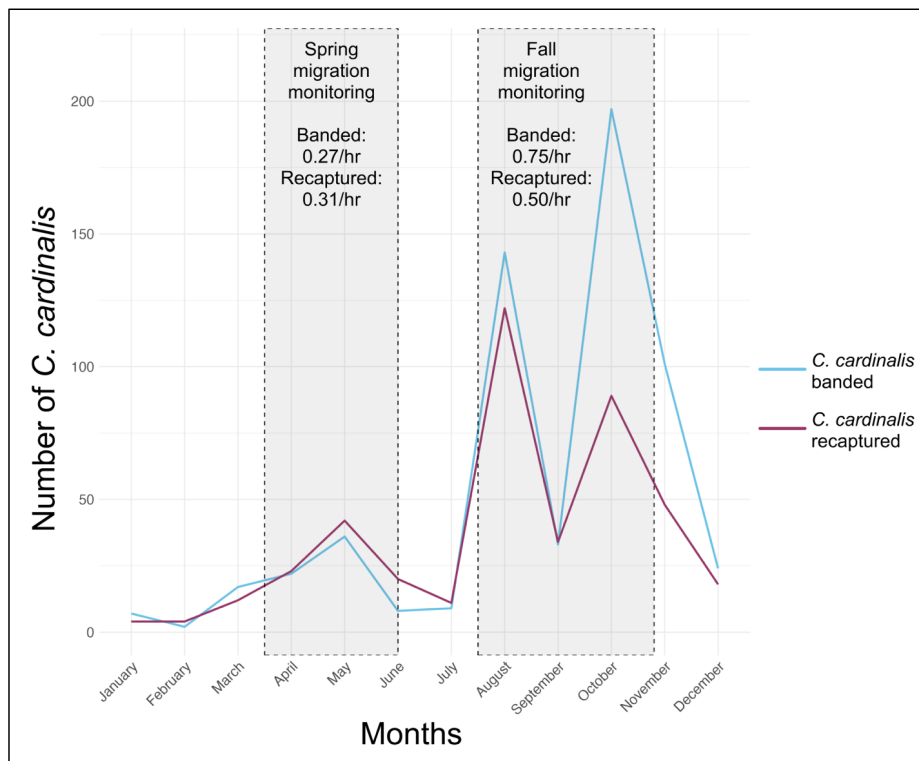


Figure 3. Number of *C. cardinalis* banded and recaptured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada. The grey boxes highlight the duration of the spring and fall migration monitoring programs, along with the approximate number of *C. cardinalis* banded and recaptured per hours of mist-net being opened during the season.

Telemetry tracking of *C. cardinalis* across an urban gradient

Across three consecutive winters, we were able to track multiple individuals at all three sites and gather valuable information regarding *C. cardinalis*' survival through the study. Table 1 details

the number of *C. cardinalis* we were able to track for each field season, their apparent survival, along with the number of confirmed mortalities and the potential causes for these deaths. cardinals' survival through the study (Table 1).

Table 1. Number of *C. cardinalis* tracked 1 and 3 months after initial release, with apparent survival (48.15% average) and fated survival (including individuals presumed alive but outside the study area; 71% average), including known mortality events/probable causes across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of <i>C. cardinalis</i>	# tracked 1 month (apparent survival)	# tracked 3 months (apparent survival)	Missing individuals after 3 months	Confirmed mortality events and probable cause of death	Survival of missing individuals (fated survival)
Peri-urban	41	34 (83%)	28 (68%)	13	6 Predation (n = 4) Unknown (n = 2)	54%
Suburban	31	19 (61%)	18 (58%)	13	4 Predation (n = 2) Disease (n = 1) Unknown (n = 1)	69%
Urban	36	11 (31%)	6 (17%)	30	3 Predation (n = 1) Window collision (n = 2)	90%

Telemetry tracking was conducted 2 to 3 days per week by three people between October and May 2022-2024, for a total of ~288 days of tracking per winter. We recorded a total of 688 unique detections were recorded at the peri-urban site, 411 at the suburban site, and 265 at the urban site. The tracking team covered the maximum amount of ground possible during each outing, to record as many individuals as feasible.

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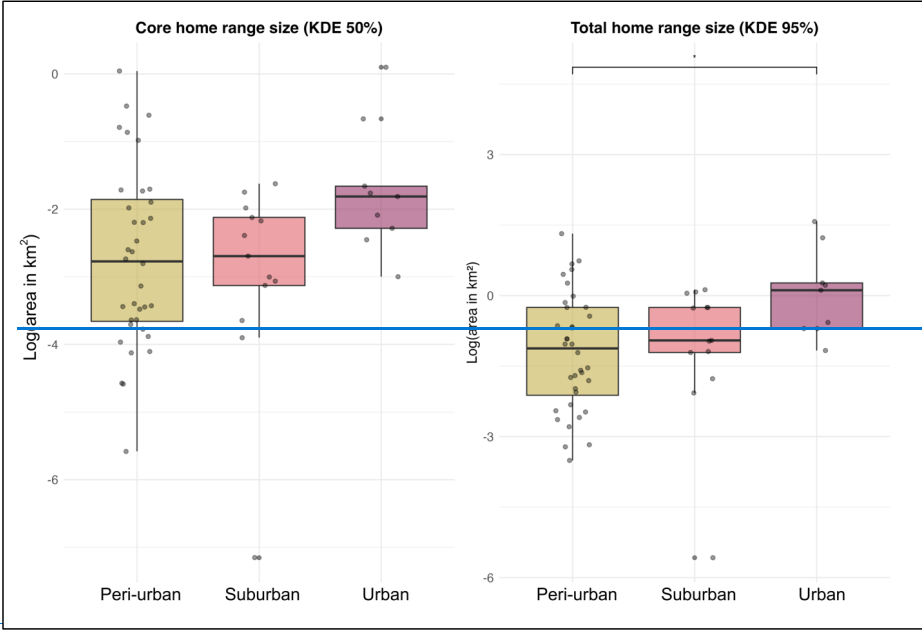
Core areas and home ranges of *C. cardinaliscardinals* across an urban gradient

We estimated core areas and home ranges for individuals with ≥ 5 recorded GPS locations (35 individuals from the peri-urban site, 13 from the suburban site and 9 from the urban site; Figure 4). Among the 58 individuals with ≥ 5 recorded GPS locations, core and total home range sizes were not normally distributed (Shapiro-Wilk tests: core, $W = 0.624$, $p < 0.001$; total, $W = 0.698$, $p < 0.001$), so areas were transformed using a logarithm prior to analysis (Figure 4; Table 2).

Table 2. Number of *C. cardinaliscardinals* that could be included in the analyses along with their respective core areas and home ranges including standard errors across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of <i>C. cardinaliscardinals</i> included in the analyses	Core area (km ²)	SE of the core area (km ²)	Home range (km ²)	SE of the home range (km ²)
Peri-urban	36	0.15	0.04	0.61	0.13
Suburban	13	0.08	0.02	0.56	0.11
Urban	9	0.28	0.11	1.53	0.52

After transformation, the data met assumptions of normality and variances were homogeneous. There was no significant difference in core home range size among sites, although the effect approached significance (ANOVA: $F_{2,55} = 2.69$, $p = 0.077$). Post-hoc comparisons show a trend towards the *C. cardinaliscardinals* at our urban site to have larger core home range sizes compared to the suburban and peri-urban sites (Figure 4). There was a significant difference between among the three study sites regarding total home ranges (ANOVA: $F_{2,55} = 3.45$, $p = 0.04$). Post-hoc TukeyHSD comparisons showed that the only significant difference was between the total home ranges of *C. cardinaliscardinals* at our urban site being larger than *C. cardinaliscardinals* at the peri-urban site (mean difference = 1.24, 95% CI [0.09, 2.38], $p = 0.03$).



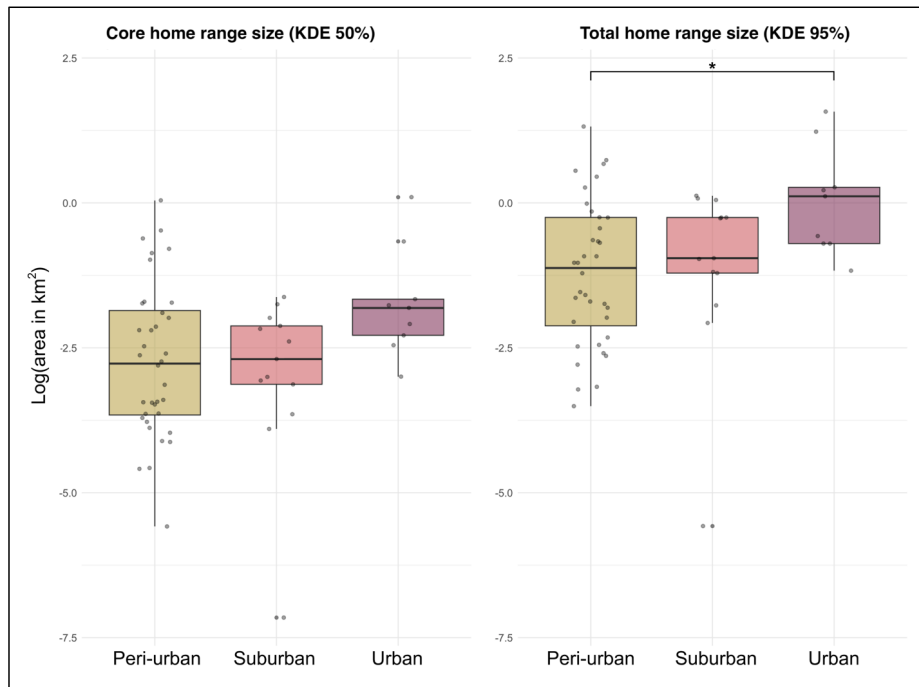


Figure 4. Comparison of the core (KDE 50%) and total (KDE 95%) home ranges of *C. cardinalis* in three study sites (peri-urban, suburban and urban) in Montreal, Quebec, Canada during winters between the years 2022 and 2025. The area in km² was log transformed, and each data point corresponds to an individual's home range (area log transformed).

Scale of association to habitat

The spatial scale where habitat variables best predicted *C. cardinalis* habitat use (i.e., the scale of effect) on the smaller side for all three sites. At the peri-urban site, the 50-meter spatial-scale model explained the most variation seen in the data compared to the other spatial-scales (i.e., > 4 A-AICc values between all other models; Figure 5A). For the birds at the suburban site and the

urban site, the best model is the 25-meter one ($> 10 \Delta AICc$ values between all other models; Figure 5B). Lastly, for the urban site, the 25-meter model was the top model from the spatial scales investigated ($> 10 \Delta AICc$ values among all other models; Figure 5C).

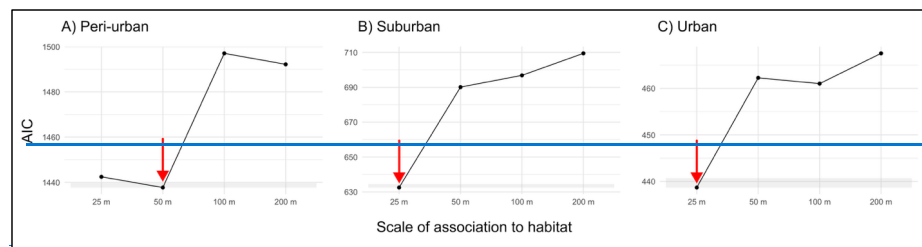


Figure 5. Model ranking using AICc of generalized linear mixed models (GLM) with a comparing scale of effect of landscape variables for resighting locations ($n=57$) of *C. cardinalis* tracked across three sites (A, B and C) in Montreal, Quebec, Canada. Arrows indicate the model with the lowest AIC value and the shaded area represent a two $\Delta AICc$ of 2 with the best model. We compared AICc among models fit with landscape covariates measured at four different spatial scales (25, 50, 100 and 200 m radius around resighting locations). At the peri-urban site, the 50 m scale model best explained cardinal's movements ($\Delta AICc > 4$ relative to all other scales). At both the suburban and urban sites, the 25 m scale model performed best ($\Delta AICc > 10$ relative to all other scales). Because the landscape variables we measured at different spatial scales are inherently correlated, these results indicate the scale at which habitat structure shows the strongest statistical association with cardinal's movements, rather than reflecting a direct causal scale of effect.

Resource selection: used vs. available habitats

Due to differences of urbanization level and scales of effect, resource selection function models were run separately for each study site. For the peri-urban site, the global model explained the

most variation seen in the data ($> 10 \Delta \text{AICc}$ values between all other models, Supplementary Material Table S1). [The full model reduced deviance by ~11.3% relative to the null model.](#) For the suburban site, the top model was the [humananthropogenic](#) covariates model (human population and road density), with the next best model (full model) $< 2\text{AICc}$ units from the top model (Supplementary Material Table S1). [The anthropogenic covariates model reduced deviance by ~16.6% relative to the null model.](#) Lastly for the urban site, the full model was the top model, but again the next best ([humananthropogenic](#) covariates) was $< 1 \text{AICc}$ from the top model (Supplementary Material Table S1). [The full model reduced deviance by ~10.5% relative to the null model.](#) The best model for each study site was then used to generate prediction plots on the probability of presence of [C. cardinaliscardinals](#) according to the model variables (Table 3).

Table 3. Model parameters for the three resource selection models predicting the presence/absence of 56 [C. cardinaliscardinals](#) within (25 or 50 m-radius) areas based on the average vegetation height (in meters) and its standard deviation, along with human-related environmental features such as population density (in people/km²), road length (in meters) and distance to the nearest bird feeder (in meters). [The data was scaled and centered before running the models.](#) The study site, year ([winter](#)) of sampling ([Winter](#)) and the individual's identification (ID) are included as [random effects in each model.](#)

Study Site	Variable	Coefficient estimate	Standard error	z	p
Peri-Urban	<i>Global model – 50 meters</i>				
	Mean vegetation height (m)	-1.272	0.153	-8.329	<0.001
	Standard deviation of the mean vegetation height	1.107	0.114	9.709	<0.001
	Mean population density (person/km ²)	-0.095	0.111	-0.858	0.391
	Road length (m)	0.023	0.110	0.207	0.836
	Distance to the nearest bird feeder (m)	-0.347	0.068	-5.067	<0.001
Suburban	HumanAnthropogenic covariates model – 25 meters	-0.486	0.135	-3.603	<0.001

Urban	Mean population density (person/km ²)	1.238	0.136	9.138	<0.001
	Road length (m)	-0.427	0.128	-3.342	<0.001
	Distance to the nearest bird feeder (m)				
	<i>Global model – 25 meters</i>				
	Mean vegetation height (m)	0.252	0.180	1.399	0.162
	Standard deviation of the mean vegetation height	-0.458	0.191	-2.405	0.016
	Mean population density (person/km ²)	-0.212	0.131	-1.620	0.105
	Road length (m)	0.736	0.135	5.438	<0.001
	Distance to the nearest bird feeder (m)	-0.392	0.141	-2.772	0.006

The prediction plots including only the significant fixed effects for the resource selection functions of each locations reveal that the presence of *C. cardinaliscardinals* is associated with different urban and habitat features according to the site they spend their winter in (Figure 65). In the peri-urban site, the probability of presence of the *C. cardinaliscardinals* decreases as the mean vegetation height increases (Figure 6A5A), while the presence of the birds increases with greater variability in vegetation height (Figure 6B5B). The probability of presence of *C. cardinaliscardinals* also seems to decrease as the distance to the nearest bird feeder increases (Figure 6C5C). In our suburban site, the probability of presence of *C. cardinaliscardinals* shows little relationship with population density (Figure 6D5D), but increases as the mean road length increases (Figure 6E5E). Like our peri-urban site, the presence of the birds also decreases as the distance to the nearest feeder increases at the suburban site (Figure 6F5F). At the urban site, *C. cardinalis'cardinals'* probability of presence decreases with increasing vegetation height standard deviation, although uncertainty is high (Figure 6G5G). The presence of *C. cardinaliscardinals* also increased with mean road length and decreased with increasing distance to the nearest feeders. Across sites, proximity to feeders and structural complexity of the infrastructures or the vegetation consistently influenced the probability of presence, yet specific relationships varied. Taken together, these results indicate that winter habitat association by cardinals varies across the urbanization gradient. While feeder availability appears to be consistently associated with a higher probability of occurrence across all sites, the habitat features associated with cardinal

presence differed among the peri-urban, suburban and urban sites. This suggests that patterns of habitat use are largely context dependent.

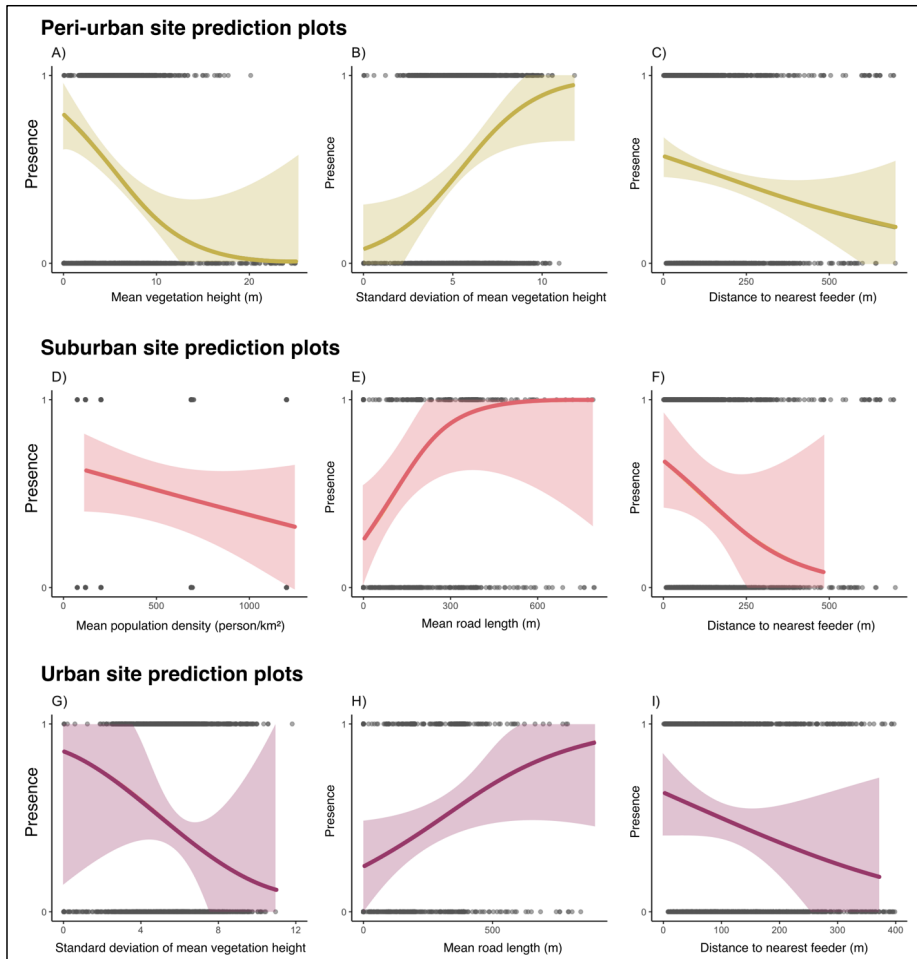


Figure 65. Probability of presence of *C. cardinaliscardinals* in a given radius area, as predicted by a generalized linear mixed effects model. Shaded areas represent 67% confidence intervals

522 around the predicted probabilities. The model's fixed effect variables are described in Table 2,
523 and for each model, random effects were sampling years, translocation site and bird ID.

524

525 **Discussion**

526 [This study examined winter space use, habitat associations and apparent survival of cardinals](#)
527 [across an urban gradient. Although habitat associations differed among sites, broad patterns](#)
528 [emerged. Individuals relocated at the urban site exhibited larger home ranges and lower apparent](#)
529 [survival, than their counterparts at the suburban and peri-urban sites, while associations with](#)
530 [environmental variables varied across the urbanization gradient. Proximity to bird feeders was](#)
531 [consistently associated with cardinal occurrence across all sites, and individuals at the suburban](#)
532 [and urban sites were primarily resighted in private UGS rather than larger public UGS. These](#)
533 [findings suggest that winter habitat use by cardinals may vary according to local habitat](#)
534 [conditions and resource availability, while anthropogenic food sources appear to be associated](#)
535 [with space use across our range of urban environments. More broadly, our results indicate that](#)
536 [private UGS may represent an important component of the habitat usage of the cardinals during](#)
537 [the non-breeding season in southern Quebec.](#)

538 [Larger home ranges at urban site and the importance of bird feeders](#)

539 [Individuals at the urban site had the largest total \(KDE 95%\) home ranges \(although we did not](#)
540 [have a truly rural site, which may have also had a large home range\). Because we couldn't](#)
541 [capture the full spectrum of urbanization, the relationship to territory size may not be linear \(i.e.,](#)
542 [larger home ranges in rural and urban areas compared to smaller home ranges in peri-urban and](#)
543 [suburban areas\).](#) *P. domesticus* in urban landscapes occupied smaller home ranges compared to
544 [their rural counterparts through the breeding season \(Vangestel et al. 2010\), O'Donnell \(2020\)](#)

also suggested that birds in urban environments generally tend to have smaller home ranges. However, female cardinals venture out of their territory more often when they had supplemental feeding, compared to females that didn't have access to supplemental food, leading to a trend towards larger home ranges (Humbird and Neudorf 2008). In our case, we found the opposite trend. By mapping the location of visible bird feeders, we obtained a coarse estimate of feeder density at all sites: the peri-urban site had ~13 feeders/km², the suburban site had ~35 feeders/km² and the urban site had ~12 feeders/km². Although these estimates provide useful context for interpreting site-level differences, they are not representative of all the feeders available and adequately maintained for the cardinals and should therefore be interpreted cautiously. One possible explanation is that the lower feeder density at the urban site, coupled with a potentially more challenging urban matrix, may have been associated with more extensive movements between reliable resource points. Because feeder density was not directly incorporated into our home range analyses, this interpretation should be considered as being on the speculative side. Conversely, the higher feeder density observed at the suburban site might have reduced the distance needed to move among reliable food resources. However, we want to point out that our study was not designed to test this mechanism directly. Future studies that are explicitly quantifying feeder availability and incorporating it into movement analyses would be valuable to evaluate whether variation in feeder density in urban areas is associated with winter home range size.

Lower apparent survival at urban site

The northern range expansion of *C. cardinalis* is largely attributed to increasing winter temperatures along with an increase in bird-feeding, allowing *C. cardinalis* to remain in the northern part of their distribution range year-round (Robb et al. 2008; Robert et al. 2019; Halkin

et al. 2021). The importance of bird-feeding along with parkland including varying vegetation structure was also apparent in our tracking data. Across three years of telemetry tracking data, *C. cardinalis* in urban locations had larger home ranges compared to those in suburban and peri-urban sites. Comparatively, the use of a fine spatial scale of habitat association (25–50 m) across the urban gradient shows that habitat use is localized in urbanized areas, highlighting the importance of having quality habitat at multiple spatial scale. What we have found on survival also indicates that urban areas are less suitable wintering locations compared to the peri-urban or suburban areas. *C. cardinalis* in suburban and urban sites do not spend time in parks or other larger public UGS during the non-breeding season, instead favouring the private UGS such as backyards and gardens and are strongly associated with the presence of anthropogenic food resources (i.e., birdfeeders). This highlights the often overlooked importance of private UGS in conservation, even though they are a dominant and highly diverse part of nature in cities (Goddard et al. 2010; Hutt-Taylor and Ziter 2022).

Lower apparent survival at urban site

While the apparent survival of *C. cardinaliscardinals* was quite similar between the peri-urban site (on average 65% after 3 months) and the suburban site (on average 57% after 3 months), it was much lower at the urban location (on average 16% after 3 months). Our apparent survival rates (which may have been influenced by emigration, which we could not account for) for the peri-urban and suburban locations are consistent with previously established annual survival in Ohio ($50.5\% \pm 10.3$ SE to $78.3\% \pm 7.6$ SE) and in Washington D.C. (between 40 to 70%) in the United States (Shustack 2008; Evans et al. 2015). However, the very-low survival rates that we are seeing in the urban location is well below what was appear notably lower than rates previously found in reported for urban habitats. *C. cardinalis*, though direct comparisons are

591 [complicated by differences in study design, season and the definition given to apparent survival](#)
 592 [across studies. Cardinals'](#) annual apparent survival was even greater as the amount of impervious
 593 surfaces (used as indicator of urbanization) increases from low to moderate [in Washington D.C.](#)
 594 (Evans et al. 2015). The lower survival in the urban environment is likely attributable to a higher
 595 concentration of anthropogenic threats; it was the only site with confirmed window collisions
 596 and indeed 67% of confirmed mortality were window collisions. Urban areas present multiple
 597 mortality risks for birds, including window collisions, strikes by vehicle and predation by
 598 domestic cats ([Loss et al. 2014; Ibáñez-Álamo et al. 2020; Lopes et al. 2024](#))([Loss et al. 2014;](#)
 599 [Ibáñez-Álamo et al. 2020; Lopes et al. 2024](#)). ~~All of which can be found~~These risks also apply in
 600 ~~both the suburban and urban site~~areas, but to a ~~different degree~~lesser extent. Specifically, our
 601 urban site had a far higher density of windows than our other sites, with many multi-level low-
 602 rise buildings (whereas none at other sites) ~~that are especially dangerous for birds, which we~~
 603 [hypothesise could have contributed to elevated collision risk](#). An alternative explanation is that
 604 the individuals translocated to the urban site survived but moved ~~further~~farther away from their
 605 release site and were beyond the area monitored (> 18 km²). ~~Nevertheless, we suggest that the~~
 606 ~~heightened~~We cannot rule out this explanation, and both processes—true mortality ~~from the~~
 607 ~~urban-specific threats is the primary driver of the reduced and undetected emigration—may have~~
 608 [contributed to the low](#) apparent survival ~~we observed. We also cannot exclude the possibility that~~
 609 ~~the *C. cardinalis* at our urban site. Given the confirmed window collisions and the higher density~~
 610 [of anthropogenic hazards, we tentatively favour a mortality driven interpretation but](#)
 611 [acknowledge this remains uncertain. A further hypothesis is that cardinals](#) faced higher risks by
 612 moving in a larger area in the urban habitat, a matrix comprised of many threats ~~like~~[including](#)

domestic cat predation ~~in addition to,~~ window ~~collisions,~~ and vehicle ~~collisions~~ ~~strikes~~ in comparison to the suburban and peri-urban areas.

~~Larger home ranges at urban site and the importance of bird feeders~~

~~Individuals at the urban site had the largest total (KDE 95%) home ranges (although we did not have a truly rural site, which may have also had a large home range). Because we couldn't capture the full spectrum of urbanization, the relationship to territory size may not be linear (i.e., larger home ranges in rural and urban areas compared to smaller home ranges in peri-urban and suburban areas). *P. domesticus* in urban landscapes occupied smaller home ranges compared to their rural counterparts (Vangestel et al. 2010). O'Donnell (2020) also suggested that birds in urban environments generally tend to have smaller home ranges. However, female *C. cardinalis* venture out of their territory more often when they had supplemental feeding, compared to females that didn't have access to supplemental food, leading to a trend towards larger home ranges (Humbird and Neudorf 2008). In our case, we found the opposite trend. By mapping the location of visible bird feeders, we determined the approximate feeder density at all sites: the peri-urban site had ~13 feeders/km², the suburban site had ~35 feeders/km² and the urban site had ~12 feeders/km². We can hypothesise that the poorer density of feeders at the urban site, along with a more dangerous matrix led *C. cardinalis*'s movements to be more convoluted and pushed them to travel further between these reliable resource points because of their scattered locations. On the contrary, in the suburban site (which has a feeder density over twice that of the urban site) we can hypothesise that the high feeder density allowed the birds to have access to reliable food sources coupled with simpler and shorter travel routes.~~

Small scale of association to habitat regardless of urbanization level

C. cardinalis Cardinals were associated with habitat features at different spatial scales depending on across our study sites, with the level of urbanization of their wintering site: strongest model support occurring at 50 meters at for the peri-urban site, and 25 meters in for both our the suburban and urban sites. This While this difference is modest and should be interpreted cautiously, the overall pattern suggests that as soon as human presence is more pronounced in the landscape, *C. cardinalis* rely more heavily on immediate is consistent with cardinals responding to relatively fine-grained habitat factors (possibly due to increased fragmentation and reduced availability of high-quality habitats features across all sites, regardless of urbanization level. Because telemetry locations and environmental layers contain some level of spatial uncertainty, and because habitat variables measured at broader adjacent scales) are correlated to some level, the estimated scale for the strongest association should be interpreted as an approximation rather than a precise measure of the scale at which cardinals select habitat features. In particular, the spatial error associated with telemetry locations (10-15 m) approaches the difference between our finest scales (25-50 m), further limiting our ability to distinguish between habitat association at these adjacent scales. This aligns with broader research suggesting that the magnitude and direction of species' responses to urbanization reflects landscape-scale differences within cities (Fidino et al. 2021). Moreover, recent work on migratory birds passing through urban areas via stopover sites highlighted the value of multiscale habitat assessment since different processes can come into play at different spatial resolution (Guo et al. 2023). Our findings showing that *C. cardinalis* respond to finer spatial scales in more urbanized areas also support evidence that UGS size and configuration matter—especially in cities where small patches may play a critical role to support bird populations. Although our study design did not test this directly, the slightly smaller

657 [scale of the strongest model supported for the suburban and urban sites is consistent with the idea](#)
 658 [that habitat configuration may influence habitat association in more urbanized areas. More](#)
 659 [broadly, our data suggest that habitat associations were detected at a relatively fine spatial scale](#)
 660 [across the urban gradient, which seems to be consistent with previous work indicating that small](#)
 661 [habitat patches could contribute and support bird communities in urban areas](#) (Dale 2018).
 662 Nineteen years of banding data at the peri-urban site demonstrated that the proportion of
 663 unbanded to banded birds was similar over most of the year, implying a mostly resident
 664 population, yet there was a distinct peak of new birds in late fall (Fig. 3S3), suggesting a local,
 665 'very short-distance' migration (including post-fledging dispersal) for *C. cardinalis*.
 666 Our findings [underscore highlight the importance potential value of scale-sensitive, species-](#)
 667 [specific approaches considering multiple spatial scales](#) when evaluating habitat use in urban
 668 areas, and [join other researchers in are consistent with previous studies](#) suggesting that
 669 [conservation efforts in urbanized environments may need to focus on fine-scale habitat quality to](#)
 670 [support characteristics may be relevant to resident species year-round in urbanized habitats](#)
 671 [\(Litteral and Shochat 2017; Guo et al. 2023\)](#) (Litteral and Shochat 2017; Guo et al. 2023). In our
 672 case, bird feeders and vegetation structure had an important impact from the smallest scales
 673 (third order habitat selection) to the entire range (first order habitat selection).
 674 [Variation of preferred habitat factors associations across an urbanization gradient](#)
 675 *C. cardinalis* Cardinals were associated with different environmental and anthropogenic habitat
 676 variables across an urban gradient, [suggesting indicating that ecological needs and drivers of](#)
 677 [winter survival by C. cardinalis in human-modified landscapes is context-dependent \(habitat](#)
 678 [associations are different among the peri-urban vs. suburban vs and urban sites.](#) The one
 679 consistent association across study sites was the relationship between *C. cardinalis*'

680 presence and a shorter distance to the closest bird feeder—which is indeed apparent even at the
 681 scale of the entire range (Fig. 2; Robb et al. 2008). This is unsurprising as *C. cardinalis*
 682 ~~is~~cardinals are one of the most common species at bird feeders within its range, especially during
 683 autumn and winter (Horn et al. 2014). ~~C. cardinalis~~ We also found a significant relationship with
 684 ~~another anthropogenic variable, wherein cardinals~~ were ~~also~~ more likely to be found in areas
 685 with ~~more paved~~higher road cover (~~a potential proxy for~~which may reflect higher residential
 686 ~~density~~development) in both the suburban and urban sites. ~~Similarly, C. cardinalis had better~~
 687 ~~survival odds~~ This pattern is consistent with previous work showing a positive association
 688 ~~between cardinals and urbanized landscapes~~ during the breeding season (~~between May and~~
 689 ~~August) in Washington D.C. in areas with high proportions of impervious surfaces (Evans et al.~~
 690 ~~2015). The association with the standard deviation of the vegetation height diverged among~~
 691 ~~sites (Evans et al. 2015). Comparatively, cardinal associations with natural cover differed across~~
 692 ~~our urbanization gradient~~. At the urban site, ~~C. cardinalis~~cardinals avoided the areas of highly
 693 contrasted vegetation structures, while at the peri-urban site they were more likely to occupy an
 694 area with an elevated standard deviation. These findings ~~highlight the importance of~~suggest that
 695 vegetation structure ~~in~~may be associated with winter habitat association, and ~~agrees~~use, which is
 696 consistent with a previous ~~work showing~~study in Delaware and Maryland identifying that
 697 vegetation structure is a particularly important variable in urban areas, specifically for *C.*
 698 ~~cardinalis~~cardinals during the breeding season (Duren et al. 2017). ~~Postfledging C.~~
 699 ~~cardinalis~~Post-fledging cardinals in urban areas strongly select for habitat with a complex
 700 understory vegetation (Ausprey and Rodewald 2011). ~~C. cardinalis do~~We did not exhibitfind
 701 evidence of the same ~~preference~~association in the peri-urban site ~~potentially because there was~~
 702 ~~much more habitat available. The reason for this discrepancy remains unclear, although variation~~

703 [in habitat availability](#) or ~~the site was more homogenous~~ habitat heterogeneity among sites may be
704 [contributing factors](#). The importance of vegetation architecture and composition in birds' habitat
705 selection ~~was~~ most likely scale-dependent (~~Deppe and Rotenberry 2008~~) ([Deppe and](#)
706 [Rotenberry 2008](#)). ~~Indeed, *C. cardinalis*~~ Our scale of effect analysis suggests that cardinals
707 selected their wintering habitats at a larger scale at the peri-urban location, compared to the
708 urban site. ~~At a larger scale, the~~ It is possible that scale-dependent differences in vegetation
709 ~~vertical structure may not be as important as say the species' composition, while at a smaller~~
710 ~~scale, habitat selection relies more on the structure~~ cues contribute to this pattern, although we
711 [caution that the difference between the scales \(25 m vs. 50 m\) is modest and that this](#)
712 [interpretation remains speculative without a direct test](#) of the vegetation ~~compared to its~~
713 composition [effect](#).

714 [Implications](#)

715 We highlight the importance of fine-scale habitat features and resource availability in shaping
716 winter ecology of a resident songbird along an urban gradient. The context-dependent responses
717 we observed underscore the [need for tailored, scale-sensitive conservation strategies](#) importance
718 [of considering local habitat conditions and resources availability when managing UGS for](#)
719 [resident bird species](#). In particular, the reliance on bird feeders and avoidance of large public
720 UGS suggests that private UGS may play a disproportionately important role in supporting
721 certain overwintering birds. [This highlights the often-overlooked importance of private UGS in](#)
722 [conservation, even though they are a dominant and highly diverse part of nature in cities](#)
723 [\(Goddard et al. 2010; Hutt-Taylor and Ziter 2022\)](#). [While our study species is not one of](#)
724 [conservation concern, it is culturally recognized and may serve as an important species to](#)
725 [connect people to nature in urban areas](#). As urban development intensifies, integrating

726 biodiversity-friendly practices such as preserving vertically complex vegetation and enhancing
727 connectivity between small habitat patches ~~could~~may help sustain ~~resident birds'~~ populations.
728 ~~Taken together of resident birds that rely on UGS during the winter. Together,~~ our results
729 highlight how habitat ~~selection~~association by a resident bird species is shaped by both landscape-
730 level context and fine-scale resource availability. This multi-scale perspective demonstrates the
731 importance ~~on~~of considering hierarchical habitat use in urban ecology and conservation. More
732 broadly, our findings join a growing body of work emphasizing the ecological importance of
733 urban landscape's elements and highlight the need for inclusive UGS planning that includes both
734 public and private green infrastructures.

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944 Supplemental material

945 [*C. cardinalis*Cardinals](#) tagging procedure

946 We mist netted *C. cardinaliscardinals* (males and females) at the McGill Bird Observatory
947 (MBO) during October and November of the year 2022, 2023 and 2024. The birds were
948 intercepted using 30 mm Japanese Nylon nets and placed in small cotton bags to reduce stress
949 before handling. As part of the migration monitoring being done at the MBO, we banded all *C.*
950 *cardinaliscardinals* with a numbered aluminum band issued by the Canadian Bird Banding
951 Office. The individuals were sexed, aged, weighed, and randomly assigned to one of the six
952 possible release sites consisting of translocation sites and control sites at the capture location. In
953 addition to the aluminum band, we fitted all *C. cardinaliscardinals* with a set of heat-sealed
954 coloured plastic bands, each with a unique combination of colours. This allowed visual
955 identification of the individuals without having to resort to a radio tag receiver. Each bird was
956 equipped with a Lotek NanoTag (NTQB-2) weighing 0.8 grams, using soft elastics in a figure 8
957 harness configuration (the total weight of the tag, harness, and glue was less than 3% of *C.*
958 *cardinalis'cardinals'* body weight). All the radio-tags were registered as part of the Motus
959 Wildlife Tracking System (Frei et al. 2024).

960 **Motus tower locations**

961 During the first winter, only 2 automated receiver towers from the Motus network were
962 deployed. The first tower is located at the peri-urban site (45°25'50.4"N 73°56'18.6"W), while
963 the second tower is located on top of a faculty building at McGill University's Macdonald
964 campus (45°24'28.3"N 73°56'20.3"W). For the second winter, those two towers were also used
965 for continuous data location, with the addition of a third temporary receiver tower that we
966 installed at the suburban site (45°24'39.3"N 73°55'48.3"W). Finally, during the third winter, a
967 fourth, permanent Motus tower was installed on top of a building at Concordia University's
968 Loyola campus location at the urban site (45°27'32.6"N 73°38'30.2"W).

969

970 **Detailed use of gain and signal strength when tracking**

971 The gain is a value between 0 and 100, representing the amplification of the received signal. By
972 increasing it, the tracker increases how sensitive the receiver is to incoming radio signals. Higher
973 gain values increase the ability to detect weaker signals but are also subject to background noise
974 and interference. Using a lower gain reduces noise but may result in missing weak signals. In
975 urbanized locations, the starting gain will rarely surpass 70-80 as the receiver picks up too much
976 noise and interference. The signal is a value ranging from 0 to 200 for the SRX800 and 0 to 250
977 for the SRX1200. It represents the strength of the detected transmission from a radio tag,
978 displayed in decibels (dB) on the receiver. A stronger signal indicates that the bird is closer;
979 trackers use this information to estimate distance and direction. Trackers begin their tracking
980 sessions with a gain of 80-70 to increase their chances of detecting nearby individuals. Once the
981 receiver picks up a [C. *cardinalis*'cardinals'](#) tag, the tracker moves toward the direction with the
982 strongest signal. Once a threshold of 150 dB (SRX800) or 200 dB (SRX1200) is attained, the
983 trackers then lower the gain by 10 and repeat this process until they reach the tagged individual.
984 The trackers only record a GPS point if the signal strength is 150 or 200 dB, with a gain of 40 or
985 less. While visual confirmation was difficult to achieve due to the nature of the release habitats
986 (high density of private property), prior testing with radio tags showed that we could assume the
987 bird to be within 10-15 meters when the gain was ≤ 30 and the signal strength ≥ 150 dB for the
988 SRX800 receivers (or 200 db for the SRX1200). The handheld receivers were either Lotek's
989 SRX800 or SRX1200. They were operated by setting the frequency to 166.380 MHz (Motus
990 tags' frequency for the Western hemisphere) and using both the gain and the signal strength to
991 get as close as possible to the tagged individuals. We were usually able to detect birds that were

992 within 200 meters of our location, and on some rare occasions were even able to pick up signals
993 of birds located as far as 800 meters away. We did encounter a significantly higher level of static
994 noise when tracking at the urban location, yet it was not enough to prevent us from finding birds
995 within the detection range. Another factor that could have complicated tracking efforts was the
996 occasional aggregation of multiple birds, which can make it difficult to obtain precise location
997 data. In most cases, no more than two birds were observed together. When larger groups were
998 detected, the majority typically dispersed before we could approach close enough to record a
999 GPS location. Overall, however, such aggregations were infrequent and are therefore unlikely to
1000 have significantly affected the accuracy of our tracking data.

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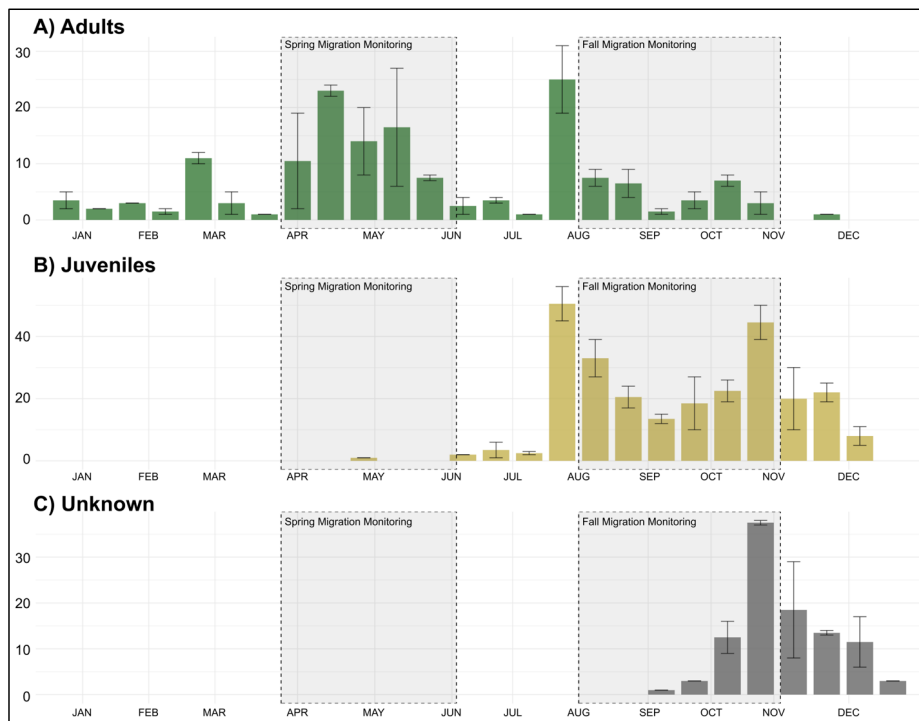


Figure S1. Mean number of *C. cardinalis* captured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada separated by age where A) shows adult birds, B) shows juvenile birds and C) shows unknown individuals.

1011 **Table S1.** Model ranking using AICc of generalized linear mixed models (GLM) with a
 1012 comparing scale-of-effect of landscape variables for resighting locations (n= 57) of *C.*
 1013 *cardinaliscardinals* tracked across three sites in Montreal, Quebec, Canada.

Study site	Model	Variables	LL	Δ AICc	Wt
Peri-urban	25 meters	Mean vegetation height + sd	-722.48	17.02	0.00
	50 meters	vegetation height + mean	-713.97	0.00	1.00
	100 meters	population density + road	-735.69	53.25	0.00
	200 meters	length + distance to nearest feeder	-740.60	46.96	0.00
Suburban	25 meters	Mean vegetation height + sd	-296.05	0.00	1.00
	50 meters	vegetation height + mean	-338.31	84.52	0.00
	100 meters	population density + road	-344.25	96.39	0.00
	200 meters	length + distance to nearest feeder	-349.92	107.73	0.00
Urban	25 meters	Mean vegetation height + sd	-211.09	0.00	1.00
	50 meters	vegetation height + mean	-225.64	29.11	0.00
	100 meters	population density + road	-224.02	25.85	0.00
	200 meters	length + distance to nearest feeder	-228.79	35.40	0.00

1014 **Table S2.** AIC values used to determine the best fitting model to explain *C.-cardinaliscardinal's*
 1015 habitat associations.

Study site	Model	Variables	AICc
Peri-urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	1443.696
		Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	1587.124
		Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	1596.077
		Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	1588.262
	Null	(1 Winter) + (1 ID) + (1 Site)	1616.101
Suburban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	631.7501
		Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	728.0640
		Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	724.0187
		Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	632.1451
	Null	(1 Winter) + (1 ID) + (1 Site)	726.1005
Urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	444.7050
		Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	486.6402
		Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	485.2496
		Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	447.8691
	Null	(1 Winter) + (1 ID) + (1 Site)	484.8853

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